॥ वसुधैव कुटुम्बकम् ॥

SYMBIOSIS INSTITUE OF TECHNOLOGY

(Established under section 3 of the UGC Act 1956)

Re-accredited by NAAC with "A" Grade

Name : Sourabh gupta

Year: AY 25-29

PRN: 25070122189

Semester: 3

ASSIGNMENT No: 5

TITLE:

Develop a Java application demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation.

Problem Statement :

Develop a Java application demonstrating the use of ArrayList, Vector, and StringBuffer for dynamic data storage, management, and string manipulation.

Theory:

ArrayList and Vector are dynamic array implementations provided by the Java Collections Framework. ArrayList offers faster performance in single-threaded applications, whereas Vector provides synchronized operations for thread safety. StringBuffer is a mutable sequence of characters that allows efficient string modification without creating new objects. These classes are commonly used in applications requiring dynamic memory allocation and efficient string processing.

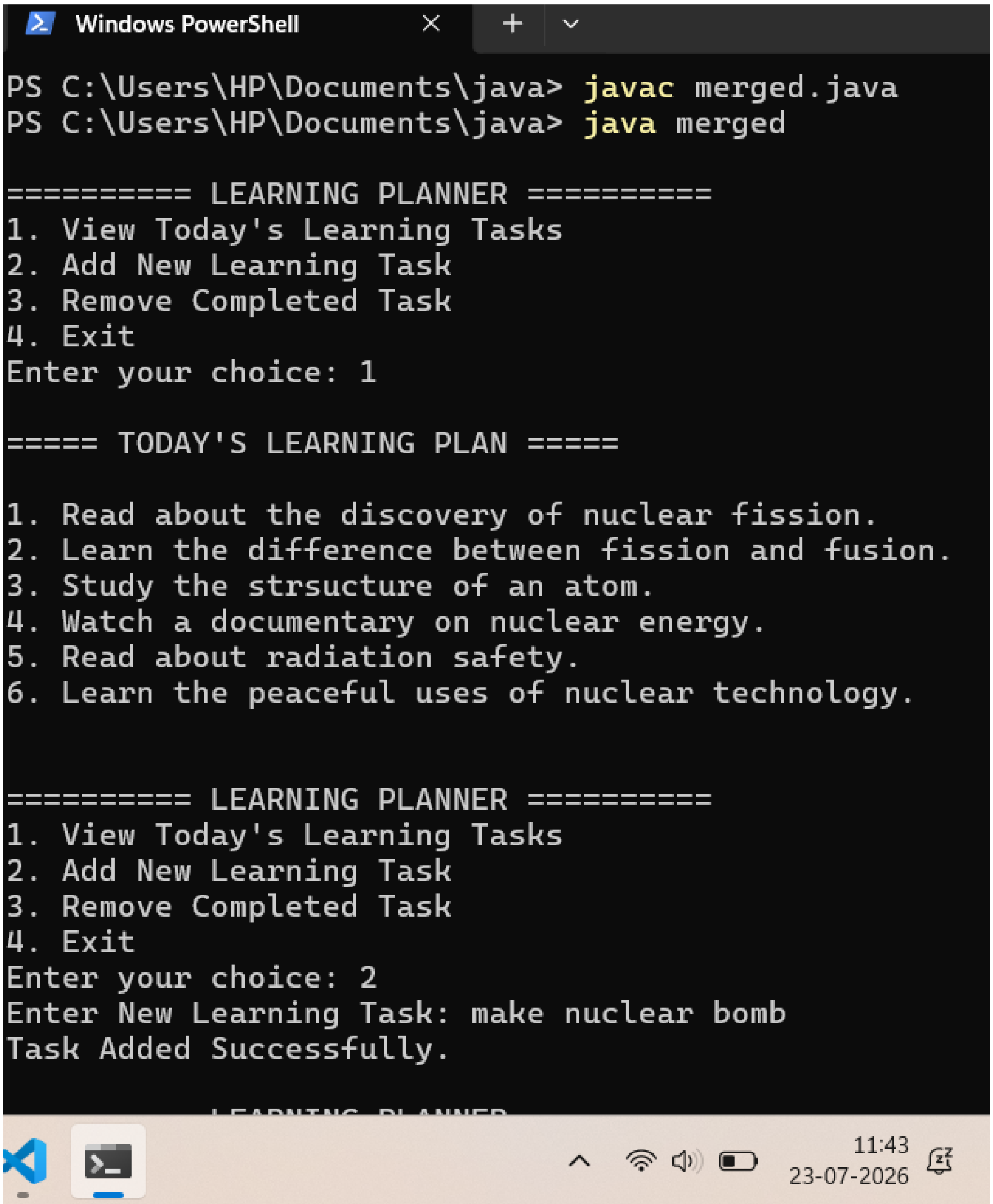
Program :

```
import java.util.ArrayList;
import java.util.Scanner;

public class merged {

    public static void main(String[] args) {
```

```
ArrayList<String> tasks = new ArrayList<>();

// Predefined learning tasks
tasks.add("Read about the discovery of nuclear fission.");
tasks.add("Learn the difference between fission and fusion.");
tasks.add("Study the strsucture of an atom.");
tasks.add("Watch a documentary on nuclear energy.");
tasks.add("Read about radiation safety.");
tasks.add("Learn the peaceful uses of nuclear technology.");

Scanner sc = new Scanner(System.in);

int choice;

do {

    System.out.println("\n========== LEARNING PLANNER ==========");
    System.out.println("1. View Today's Learning Tasks");
    System.out.println("2. Add New Learning Task");
    System.out.println("3. Remove Completed Task");
    System.out.println("4. Exit");
    System.out.print("Enter your choice: ");

    choice = sc.nextInt();
    sc.nextLine();

    switch (choice) {

        case 1:

            if (tasks.isEmpty()) {
                System.out.println("No tasks available.");
            } else {

                StringBuffer sb = new StringBuffer();

                sb.append("\n===== TODAY'S LEARNING PLAN =====\n\n");

                for (int i = 0; i < tasks.size(); i++) {
                    sb.append((i + 1) + ". " + tasks.get(i) + "\n");
                }

                System.out.println(sb);
            }

            break;

        case 2:

            System.out.print("Enter New Learning Task: ");
            String newTask = sc.nextLine();

            tasks.add(newTask);

            System.out.println("Task Added Successfully.");

            break;

        case 3:

            if (tasks.isEmpty()) {
                System.out.println("No tasks available to remove.");
            } else {

                System.out.println("\nCurrent Tasks:");

                for (int i = 0; i < tasks.size(); i++) {
```

```
                System.out.println((i + 1) + ". " + tasks.get(i));
            }
            System.out.print("Enter task number to remove: ");
            int remove = sc.nextInt();
            if (remove >= 1 && remove <= tasks.size()) {
                tasks.remove(remove - 1);
                System.out.println("Task Removed Successfully.");
            } else {
                System.out.println("Invalid Task Number.");
            }
          }
          break;
        case 4:
          System.out.println("Thank You!");
          break;
        default:
          System.out.println("Invalid Choice.");
      }
    } while (choice != 4);
    sc.close();
  }
}
```

Output :




```
PS C:\Users\HP\Documents\java> javac merged.java
PS C:\Users\HP\Documents\java> java merged

========== LEARNING PLANNER ==========
1. View Today's Learning Tasks
2. Add New Learning Task
3. Remove Completed Task
4. Exit
Enter your choice: 1

===== TODAY'S LEARNING PLAN =====

1. Read about the discovery of nuclear fission.
2. Learn the difference between fission and fusion.
3. Study the strsucture of an atom.
4. Watch a documentary on nuclear energy.
5. Read about radiation safety.
6. Learn the peaceful uses of nuclear technology.


========== LEARNING PLANNER ==========
1. View Today's Learning Tasks
2. Add New Learning Task
3. Remove Completed Task
4. Exit
Enter your choice: 2
Enter New Learning Task: make nuclear bomb
Task Added Successfully.
```

```
. Exit
nter your choice: 2
nter New Learning Task: make nuclear bomb
ask Added Successfully.

========= LEARNING PLANNER ==========
. View Today's Learning Tasks
. Add New Learning Task
. Remove Completed Task
. Exit
nter your choice: 3

urrent Tasks:
. Read about the discovery of nuclear fission.
. Learn the difference between fission and fusion.
. Study the strsucture of an atom.
. Watch a documentary on nuclear energy.
. Read about radiation safety.
. Learn the peaceful uses of nuclear technology.
. make nuclear bomb
nter task number to remove: 5
ask Removed Successfully.

========= LEARNING PLANNER ==========
. View Today's Learning Tasks
. Add New Learning Task
. Remove Completed Task
. Exit
nter your choice: 1
```

11:43
23-07-2026

```
Enter your choice: 1

===== TODAY'S LEARNING PLAN =====

1. Read about the discovery of nuclear fission.
2. Learn the difference between fission and fusion.
3. Study the strsucture of an atom.
4. Watch a documentary on nuclear energy.
5. Learn the peaceful uses of nuclear technology.
6. make nuclear bomb


========== LEARNING PLANNER ==========
```

Exercise 1

Create a To-Do List application using ArrayList to store tasks and StringBuffer to display tasks.

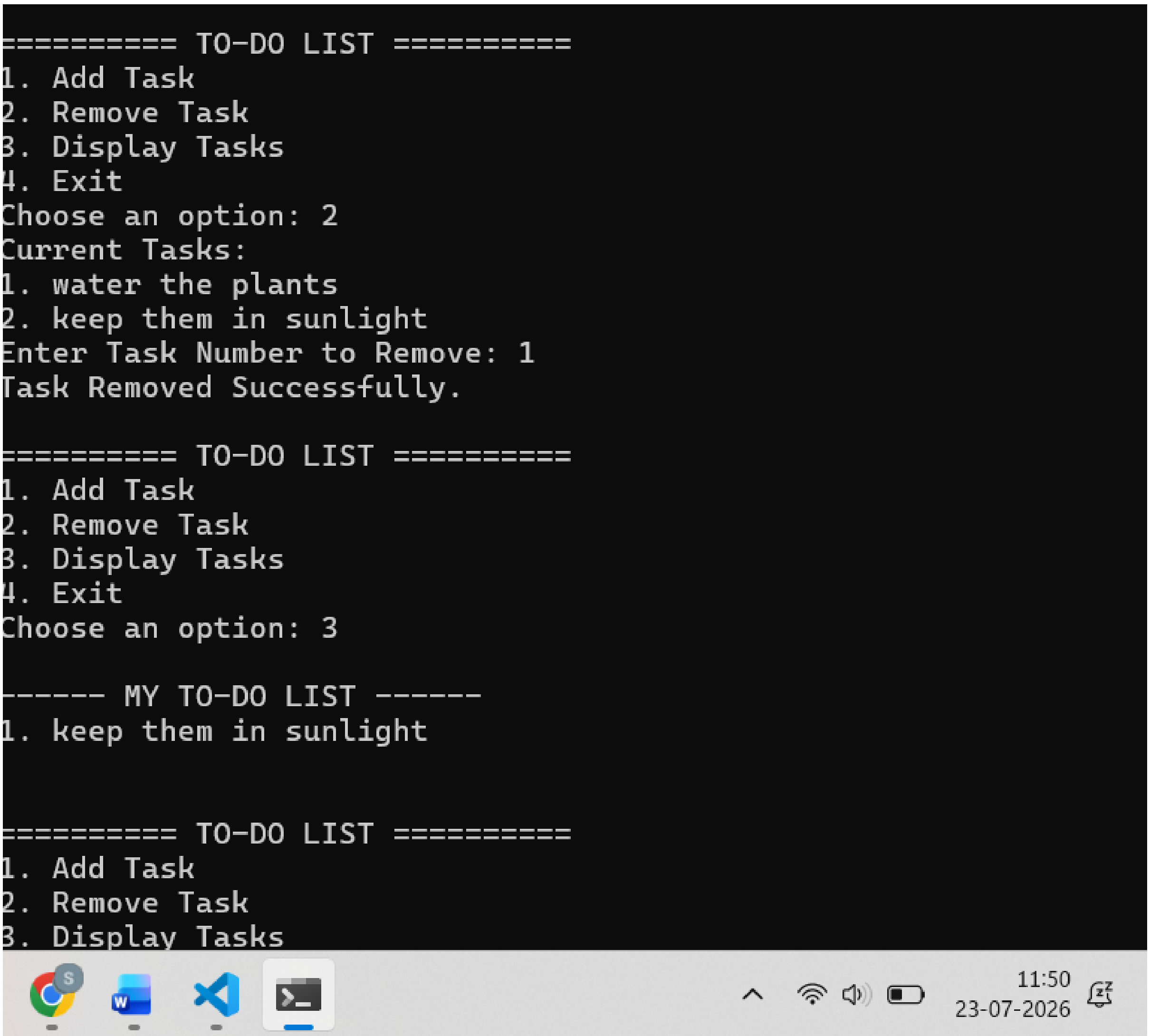
Source code:

```
import java.util.ArrayList;
import java.util.Scanner;

public class TodoList {

    public static void main(String[] args) {

        ArrayList<String> todo = new ArrayList<>();
        Scanner input = new Scanner(System.in);

        int choice;

        do {

            System.out.println("\n========== TO-DO LIST ==========");
            System.out.println("1. Add Task");
            System.out.println("2. Remove Task");
            System.out.println("3. Display Tasks");
            System.out.println("4. Exit");
            System.out.print("Choose an option: ");

            choice = input.nextInt();
            input.nextLine();

            switch(choice) {

                case 1:

                    System.out.print("Enter Task: ");
                    String task = input.nextLine();

                    todo.add(task);

                    System.out.println("Task Added Successfully.");
                    break;
```

```
case 2:
    if(todo.isEmpty()) {
        System.out.println("Task List is Empty.");
    } else {
        System.out.println("Current Tasks:");
        for(int i=0;i<todo.size();i++) {
            System.out.println((i+1)+". "+todo.get(i));
        }
        System.out.print("Enter Task Number to Remove: ");
        int remove = input.nextInt();
        if(remove>=1 && remove<=todo.size()) {
            todo.remove(remove-1);
            System.out.println("Task Removed Successfully.");
        }
        else {
            System.out.println("Invalid Task Number.");
        }
    }
    break;
case 3:
    StringBuffer sb = new StringBuffer();
    sb.append("\n------ MY TO-DO LIST ------\n");
```

```
                if(todo.isEmpty()) {

                    sb.append("No Tasks Available.\n");

                }

                else {

                    for(int i=0;i<todo.size();i++) {

                        sb.append((i+1)+". "+todo.get(i)+"\n");

                    }

                }

                System.out.println(sb);

                break;

            case 4:

                System.out.println("Closing Application...");
                break;

            default:

                System.out.println("Invalid Choice.");

        }

    }while(choice!=4);

    input.close();

  }

}
```

Output:



2. **Create a Student Course Registration System using ArrayList to store the list of courses registered by a student and StringBuffer to generate and display the registered course list. The application should allow users to add, remove, and view registered courses.**

Source code:

```
import java.util.ArrayList;
import java.util.Scanner;

public class CourseRegistration {

    public static void main(String[] args) {

        ArrayList<String> registeredCourses = new ArrayList<>();
```

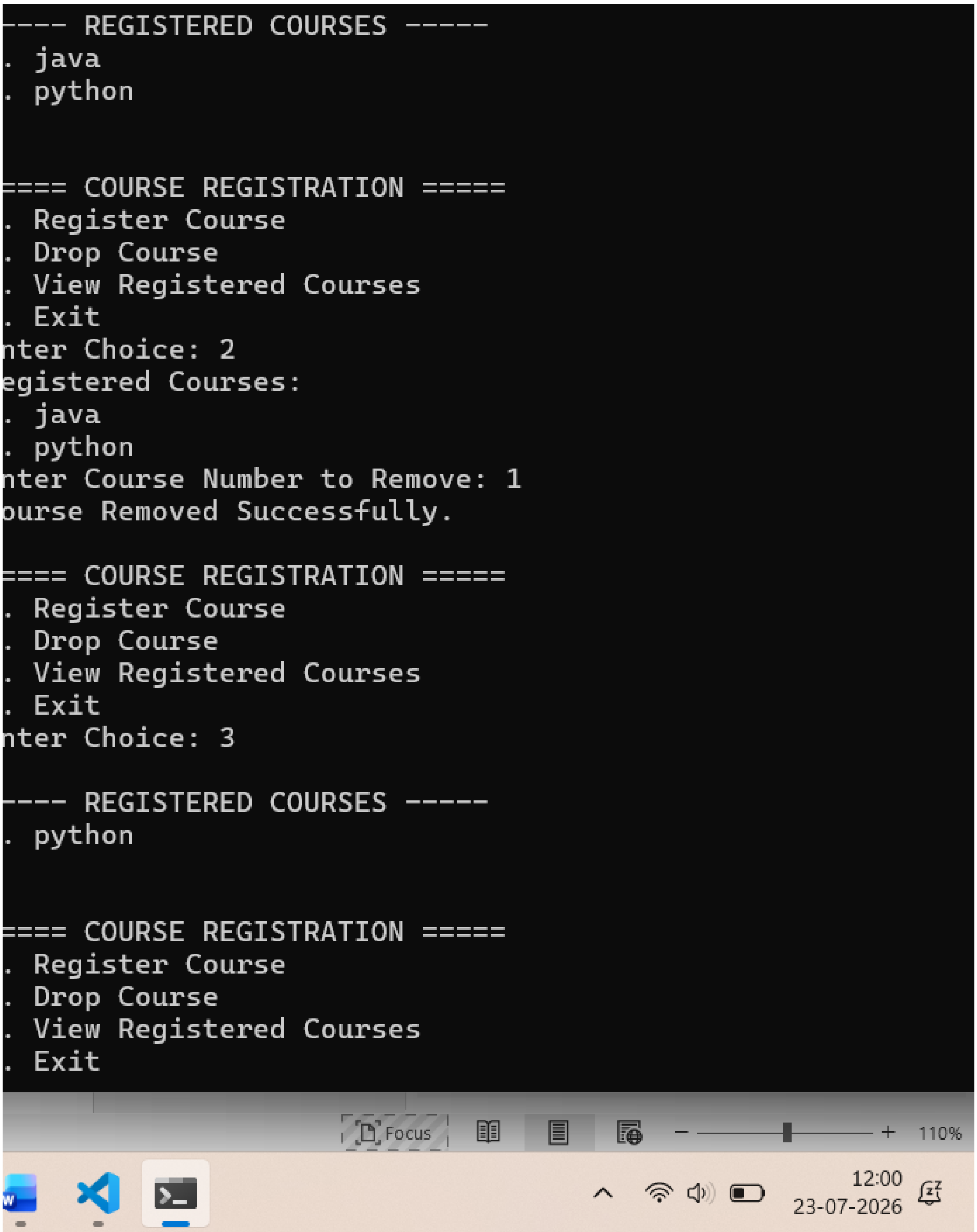
```
Scanner input = new Scanner(System.in);

int option;

do {

    System.out.println("\n===== COURSE REGISTRATION =====");
    System.out.println("1. Register Course");
    System.out.println("2. Drop Course");
    System.out.println("3. View Registered Courses");
    System.out.println("4. Exit");
    System.out.print("Enter Choice: ");

    option = input.nextInt();
    input.nextLine();

    switch(option) {

        case 1:

            System.out.print("Enter Course Name: ");

            String course = input.nextLine();

            registeredCourses.add(course);

            System.out.println("Course Registered Successfully.");

            break;

        case 2:

            if(registeredCourses.isEmpty()) {

                System.out.println("No Registered Courses.");

            }

            else {

                System.out.println("Registered Courses:");

                for(int i=0;i<registeredCourses.size();i++) {
```

```
            System.out.println((i+1)+". "+registeredCourses.get(i));
        }
        System.out.print("Enter Course Number to Remove: ");
        int remove = input.nextInt();
        if(remove>=1 && remove<=registeredCourses.size()) {
            registeredCourses.remove(remove-1);
            System.out.println("Course Removed Successfully.");
        }
        else {
            System.out.println("Invalid Choice.");
        }
    }
    break;
case 3:
    StringBuffer sb = new StringBuffer();
    sb.append("\n----- REGISTERED COURSES -----\n");
    if(registeredCourses.isEmpty()) {
        sb.append("No Courses Registered.\n");
    }
    else {
        for(int i=0;i<registeredCourses.size();i++) {
```

```
                sb.append((i+1)+". "+registeredCourses.get(i)+"\n");

              }

            }

            System.out.println(sb);

            break;

          case 4:

            System.out.println("Thank You!");
            break;

          default:

            System.out.println("Invalid Option.");

        }

      }while(option!=4);

      input.close();

  }

}
```

Output:

```
===== COURSE REGISTRATION =====
1. Register Course
2. Drop Course
3. View Registered Courses
4. Exit
Enter Choice: 1
Enter Course Name: java
Course Registered Successfully.

===== COURSE REGISTRATION =====
1. Register Course
2. Drop Course
3. View Registered Courses
4. Exit
Enter Choice: 1
Enter Course Name: python
Course Registered Successfully.

===== COURSE REGISTRATION =====
1. Register Course
2. Drop Course
3. View Registered Courses
4. Exit
Enter Choice: 3

----- REGISTERED COURSES -----
1. java
2. python
```



Conclusion :

The assignment demonstrated the use of **ArrayList** for dynamic data management and **StringBuffer** for string manipulation. It enhanced the understanding of Java collections and their application in developing efficient and flexible programs.